

CANDIDATE
NAME

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INDEX NUMBER

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BIOLOGY

9744/03

Long Structured and Free response Question
Paper 3

14 September 2017
Thursday

Candidates answer on the Question Paper.
No Additional Materials are required.

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staplers, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of a scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4 OR	
5	
Total	55

Section A

Answer **all** the questions in this section.

- 1** Cyanobacteria are a group of photosynthetic, nitrogen-fixing bacteria that live in a wide variety of moist soils and water either freely or in a symbiotic relationship with plants. Some cyanobacteria float in water by forming gas vesicles that are bounded by a protein sheath.

Fig. 1.1 below shows a generalized drawing of a cyanobacterium. The plasma membrane of cyanobacterium consists of an outer and inner membrane which is not represented in Fig. 1.1.

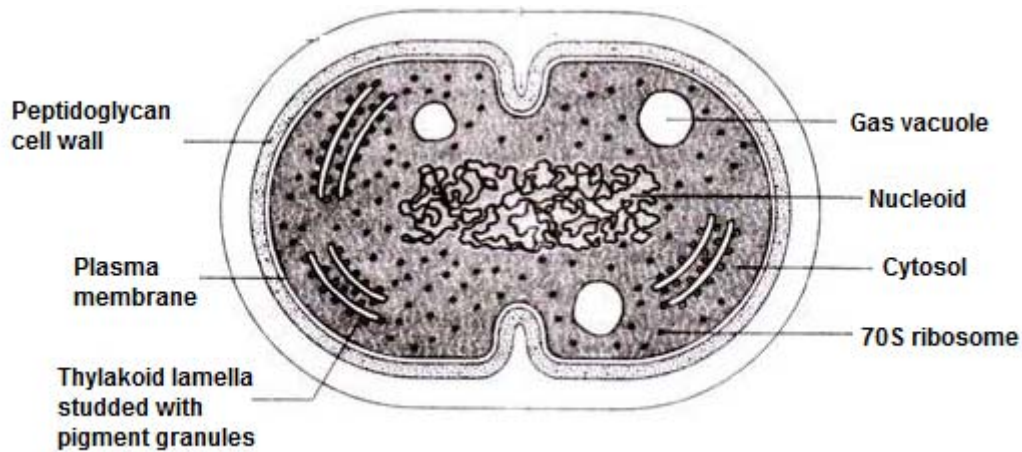


Fig. 1.1

- (a)** From Fig. 1.1, state two structural features that are expected in a typical prokaryote and two structural features that are not expected in a typical prokaryote.

Expected:

.....

.....

Not expected:

.....

.....

(b) Process of photosynthesis that occurs in cyanobacterium is largely similar to photosynthesis in chloroplast. Fig. 1.2 shows the effect of carbon dioxide concentration on the light-independent stage of photosynthesis in *Synechococcus* genus of cyanobacterium. The following steps were carried out in a study:

- a cell suspension of *Synechococcus* was illuminated using a bench lamp.
- the suspension was supplied with carbon dioxide at a concentration of 1% for 200 seconds.
- the concentration of carbon dioxide was then reduced to 0.03% for a further 200 seconds.
- the concentration of RuBP and glycerate-3-phosphate (GP) were measured at regular intervals.
- the temperature of the suspension was maintained at 25 °C throughout the investigation.

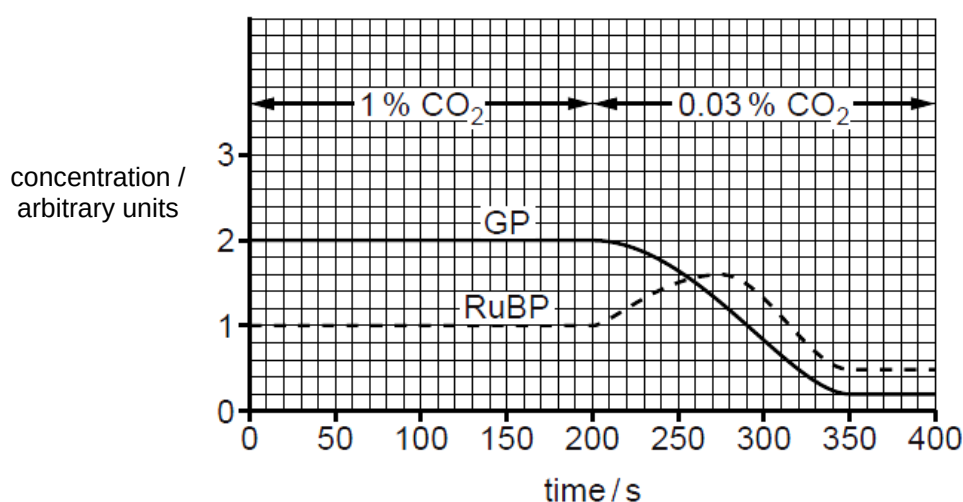


Fig. 1.2

(i) With reference to Fig. 1.2, explain why the concentration of RuBP changed between 200 and 275 seconds.

- (ii) Suggest how the decrease in the concentration of GP leads to an increase in the generation time (time it takes for the population to double) of *Synechococcus*.

[3]

- (iii) Scientists have suggested that chloroplast may have originated as cyanobacterium that continued to function after becoming engulfed by primitive eukaryotic cells, in a process similar to endocytosis.

Describe two features of chloroplast that provide support for this hypothesis.

[2]

- (c) In the study of evolution of Man, the theory of natural selection is widely used to understand how speciation of humans has occurred. The study of fossils and genetic sequences are now commonly used to help us understand more about human evolution. It is widely believed that humans are closely related to the Great Apes – chimpanzees, gorillas and orang utan, and share a common ancestor millions of years go.

Fig. 1.3 below shows some comparisons of skull structure between the Great Apes and modern Man (*Homo sapiens*).

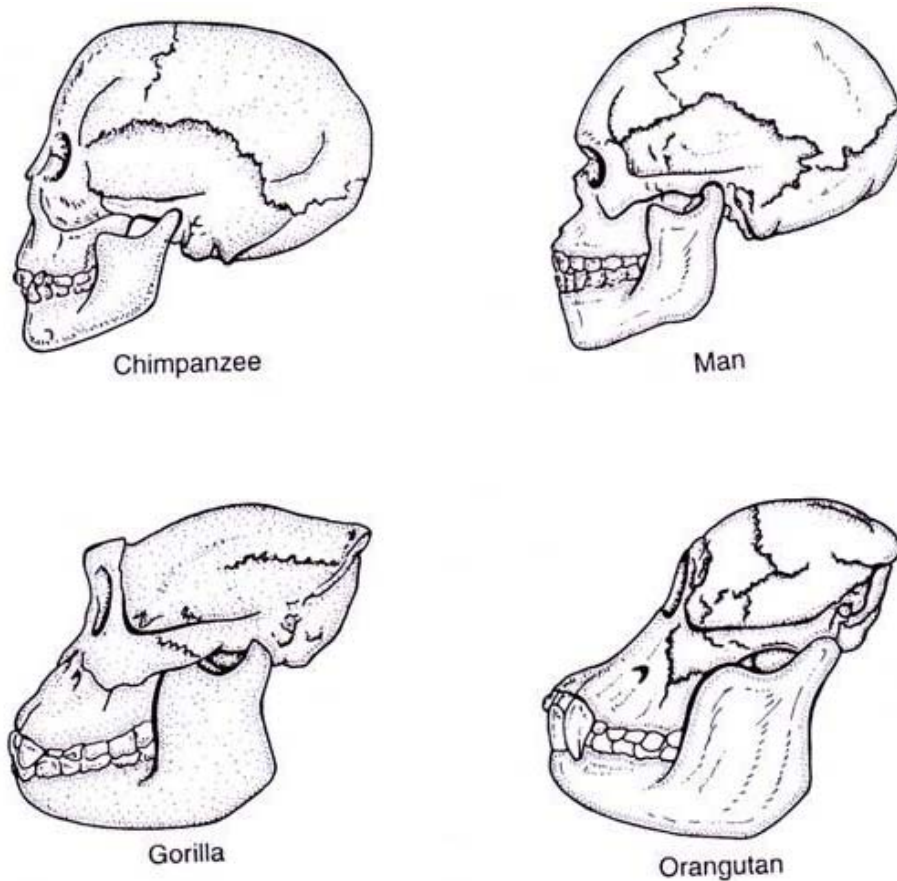


Fig. 1.3

- (i) Using Fig. 1.3, state two features that support the hypothesis that modern Man shares a common ancestor with the Great Apes.

[2]

Neanderthals (*Homo neanderthalensis*), another primate similar to modern humans is our closest human relative. Fig. 1.4 below shows a comparison between the fossilized skull of a Neanderthal and modern Man.

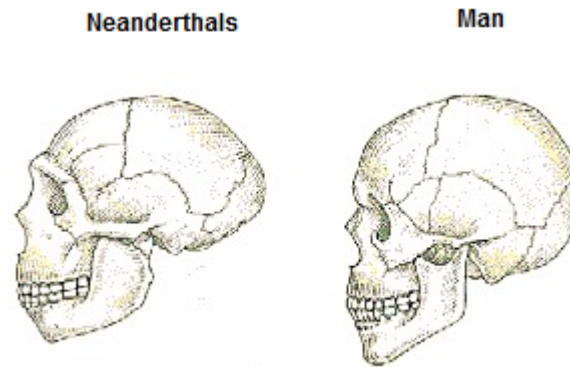


Fig. 1.4

- (ii) Disagreement exists as to whether the scientific name for Neanderthals should be *Homo sapiens* or *Homo neanderthalensis*.

With reference to **four** different species concepts, explain why it is difficult to assign a scientific name to Neanderthals.

[4]

- (iii) Discuss one advantage of using genetic sequences to study evolution of Man.

[2]

[Total: 17]

- 2 B-lymphocytes respond to the presence of a non-self antigen by dividing as shown in Fig. 2.1.

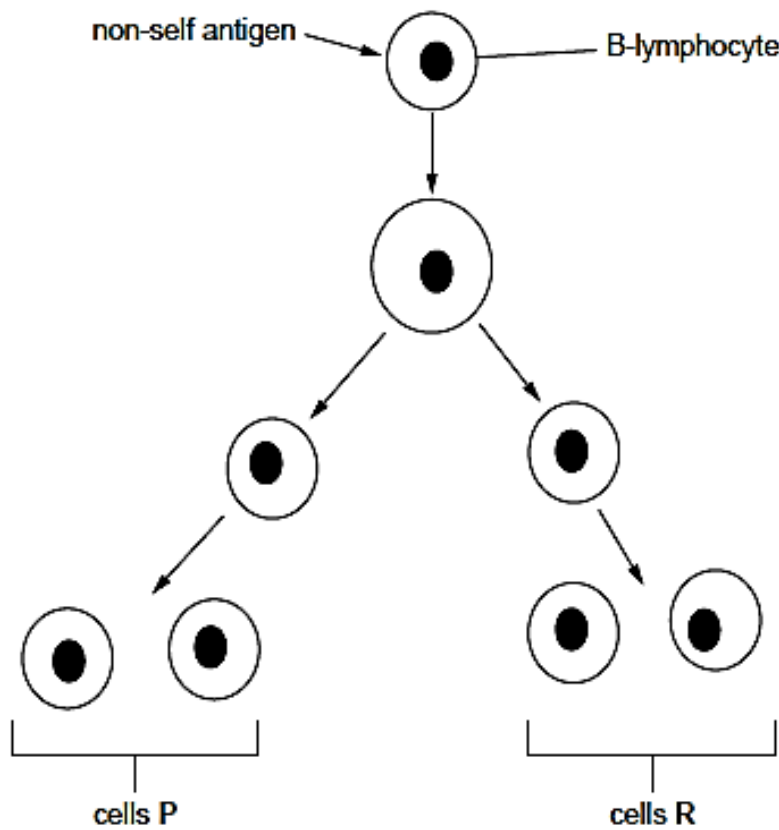


Fig. 2.1

- (a) During an immune response, cells divide by mitosis. Describe how mitosis is involved in an immune response.

- (b)** The cells labelled P on Fig. 2.1 continue to divide to give rise to many cells that differentiate into short-lived plasma cells. The plasma cells release antibody molecules.

(i) Outline how plasma cells produce antibody molecules.

[4]

(ii) Describe how antibody molecules are released from the plasma cell.

[2]

- (c)** Both B and T lymphocytes are part of adaptive immunity. Describe the mode of action of T-lymphocytes during an immune response.

[4]

- (d) Immune response is mounted against pathogen such as bacteria. Explain why phagocytes act only against the bacteria and not against human cells.

[4]

[Total: 17]

- 3 The olive tree, *Olea europaea*, is a small tree native to the Mediterranean area of Europe, Africa and parts of Asia, where it has been cultivated for several thousand years. In 1993, Beerling and Chaloner carried out estimates of stomatal density on preserved olive leaves. The oldest of these were obtained from the tomb of the Egyptian King Tutankhamun who died over 3000 years ago. The results of the study are summarised in Fig. 3.1.

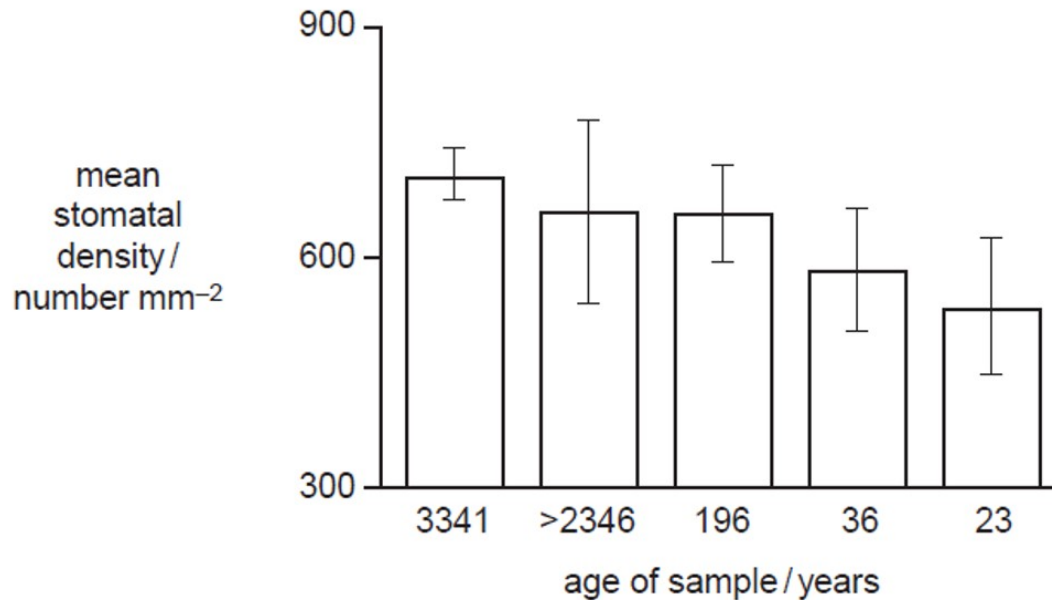


Fig. 3.1

- (a) (i) Describe the results shown by the data in Fig. 3.1.

[2]

- (ii) Explain why it is difficult to reach a valid conclusion about changes in stomatal density over time

[4]

Over the last 10 years, Kenya has made progress in malarial control. However, the country is still far from defeating the disease.

Fig. 3.2 shows how prevalence of malaria is across the country.

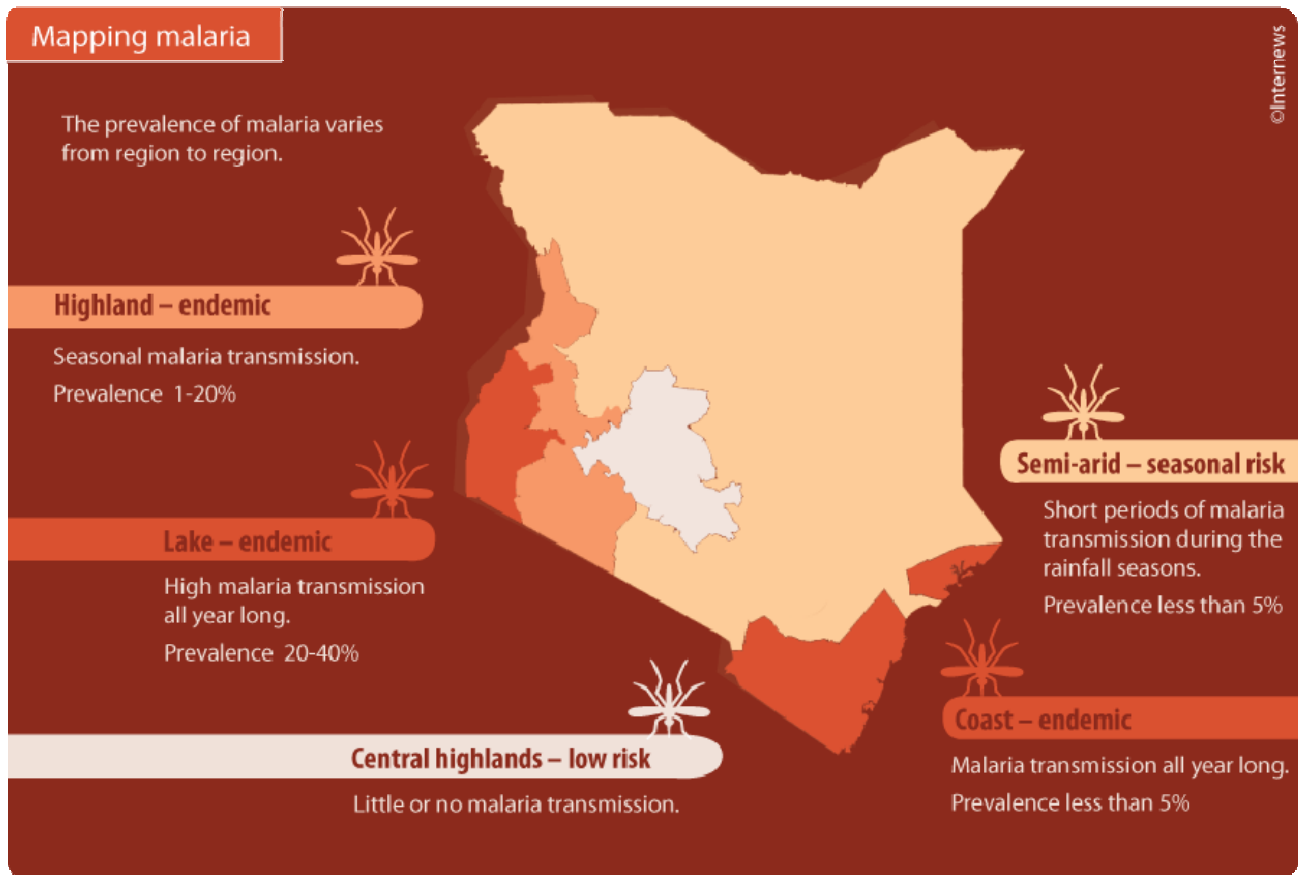


Fig. 3.2

- (b) (i) With reference to Fig. 3.2, explain the two determining factors that lead to uneven prevalence of malaria across the country.

- (ii) Suggest why it is difficult to control malaria worldwide, apart from reasons associated with global warming.

[2]

- (c) Besides mosquito-borne diseases, describe **two** other problems caused by a change in insect population as a result of climate change.

[4]

[Total: 16]

Your answer must be set out in parts **(a)**, **(b)** etc., as indicated in the question.

- [Total: 25]

[Total: 25]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

